Illustration 2.3

When state support does not help

Research by Hu and Hun (2014) illustrates that, sometimes, state support does not work. The Taiwan system of innovation in the pharmaceutical industry has failed to achieve international competitiveness, despite strong state support. Their study showed that the innovation performance of Taiwan's pharmaceutical industry has been poor when compared to India, especially when looking at patenting and publication activities. Previous research has demonstrated that patents and publications play a critical role in linking actors and institutions and is highly associated with the effectiveness of the innovation system.

Entrepreneurship

Joseph Schumpeter was one of the few intellectuals who saw business differently. He regarded business people as unsung heroes: men and women who create new enterprises through the sheer force of their wills and imaginations and, in so doing, are responsible for the most benign development in human history, the spread of mass affluence. 'Queen Elizabeth [I] owned silk stockings,' he once observed. 'The capitalist achievement does not typically consist in providing more silk stockings for queens but in bringing them within the reach of factory girls in return for steadily decreasing amounts of effort . . . The capitalist process, not by coincidence but by virtue of its mechanism, progressively raises the standard of life of the masses.' But Schumpeter recognised that business people are often ruthless robber barons, obsessed by their dreams of building 'private kingdoms' and willing to do anything to crush their rivals.

Schumpeter's ability rested on a broader philosophy of capitalism. He argued that innovation is at the heart of economic progress. It gives new businesses a chance to replace old ones, but it also dooms those new businesses to fail, unless they can keep on innovating (or find a powerful government patron). In his most famous phrase, he likened capitalism to a 'perennial gale of creative destruction'.

For Schumpeter, the people who kept this gale blowing were entrepreneurs. He was responsible for identifying the entrepreneur's central function: of moving resources, however painfully, to areas where they can be used more productively. Importantly, he also recognised that big businesses can be as innovative as small ones.

Entrepreneurship is all about stepping into the unknown and breaking away from the familiar. For some people, this is less of a challenge than for others. This can be for a wide variety of reasons, including background and parental influence. Across Europe, unlike in the USA, it remains true that more graduates would sooner work for Siemens, Shell or Nokia than start their own business. This picture, however, is changing and the numbers interested in starting their own business is rising rapidly. Furthermore, an injection of entrepreneurship, by which creative people are encouraged to strike out and develop new products or services, is important to the financial health of all organisations. Yet, fundamental questions appear as major obstacles:

Where is the opportunity? How do I capitalise on it? What resources do I need? How do I gain control over them?

For many, the entrepreneur and entrepreneurship is best captured by George Bernard Shaw's famous quote:

> *The reasonable man (woman) adapts himself (herself) to the world. The unreasonable one persists in trying to adapt the world to himself (herself). Therefore, all progress depends on unreasonable men (and women).*
>
> (George Bernard Shaw)

This captures the essence of entrepreneurship. That it is about change; it is about doing something different. Often, this change will be met with resistance and it is the entrepreneur who will persist and get things done. Without such people, improvements are less likely.[1]

But what does *entrepreneurial* mean? Managers describe entrepreneurship with such terms as innovative, flexible, dynamic, risk taking, creative and growth oriented. The popular press, on the other hand, often defines the term as starting and operating new ventures. That view is reinforced by the alluring success of such upstarts as SAP, Dyson and TomTom.

Neither approach to a definition of entrepreneurship is precise or prescriptive enough for managers who wish to be more entrepreneurial. Everybody wants to be innovative, flexible and creative. But for every SAP, Dyson and TomTom there are thousands of new restaurants, clothing stores and consulting firms that, presumably, have tried to be innovative, to grow and to show other characteristics that are entrepreneurial — but have failed.

In this book, we want to reinforce the idea that entrepreneurship is not just about starting a new business. Our focus on innovation management underscores this point. We are all aware of many medium and small businesses that consistently develop new products and markets and also grow at rates far exceeding national averages. For example, Subocean Group grew an incredible 237 per cent in 2009, increasing its turnover from £1.7 million to £65 million in 2009. Subocean runs power cables along the sea floor from offshore wind farms to substations on land (Fast Track 100, 2010). Moreover, we are all aware of many of the largest corporations – BMW, AstraZeneca and Ericsson are just a few of the best known – that make a practice of innovating, taking risks and showing creativity. And they continue to expand.

Entrepreneurship and innovation

In the United States, the subject of entrepreneurship has been taught in business schools for over 50 years. The content of these courses clearly varies but many of them study growing a small business into a large one. When it comes to innovation management, generally, this has not been studied and, until recently, there were far fewer courses available. This is changing. Yet, in Europe, we have a long history of teaching innovation management but not entrepreneurship. This is changing and entrepreneurship is a rapidly growing subject in universities across Europe, as an academic field entrepreneurship has flourished. In 1983, Babson held the first

[1] George Bernard Shaw (1856 –1950) was an Irish playwright and a co-founder of the London School of Economics. He is also the only person to be awarded a Nobel prize for literature and an Oscar.

research conference on entrepreneurship with 37 papers presented. In 2010, there were over 1,000 papers presented. Entrepreneurship now has its own division within the Academy of Management.

In Europe, there is a recognition and a considerable emphasis, especially within the technical universities, on trying to understand how entrepreneurship and innovation can help create the new technology intensive businesses of tomorrow. Moreover, it is the recognition of the entrepreneur's desire to change things that is so important within innovation. We will see later that the role of an entrepreneur is central to innovation management.

Trying to uncover separate definitions for innovation and entrepreneurship is, increasingly, a purely academic exercise. The main traits associated with entrepreneurship, such as growth, flexibility and creativity, are also desirable traits for innovation. Theorists and practioners alike recognise that these constructs are close relatives or two sides of the same coin. Now we will outline briefly the roots of these terms and their linkage.

Traditionally, it is Jean-Baptiste Say who is credited for coining the word and advancing the concept of the entrepreneur, but, in fact, it was Richard Cantillon who first introduced the term in *Essai*; this was written in 1730. Cantillon divided society into two principal classes – fixed-income wage earners and non-fixed income earners. Entrepreneurs, according to Cantillon, are non-fixed income earners who pay known costs of production, but earn uncertain incomes, hence it was Cantillon who saw the entrepreneur as a risk-taker, whereas Say predominately considered the entrepreneur a planner.

A few years later, in his 1776 thought-provoking book *The Wealth of Nations*, Adam Smith explained clearly that it was not the benevolence of the baker but self-interest that motivated him to provide bread. From Smith's standpoint, entrepreneurs were the economic agents who transformed demand into supply for profits. In 1848, the famous economist John Stuart Mill described entrepreneurship as the founding of a private enterprise. This encompassed the risk takers, the decision makers and the individuals who desire wealth by managing limited resources to create new business ventures.

Whilst entrepreneurship may have a long history, the term entrepreneur continued to be used to define a businessman until the arrival of Joseph Schumpeter. It was his work in the 1930s that made the clear linkage between the terms innovation and entrepreneurship. He considered entrepreneurship as influencing growth in the economy. It is something that disrupts the market equilibrium, or 'circular flow'. Its essence is 'innovation'. He writes that: 'the carrying out of new combinations we call enterprise; the individuals whose function is to carry them out we call entrepreneurs' (1934: 74). After Schumpter's work, most economists (and many others) have accepted his identification of entrepreneurship with innovation (see Kilby, 1971, for a summary of the term entrepreneur).

According to Schumpeter, economic development is the result of three types of factors:

1 External factors such as demand by government (changes in legislation, defence orders).
2 Factors of growth or gradual changes in economic life that are accomplished through day-to-day activities and adjustments.
3 'The outstanding fact in the economic history of capitalist society', innovation.

And, for Schumpeter, entrepreneurs are galvanised into action under the following conditions (Schumpeter, 1961: 214):

1 The existence of new possibilities more advantageous from the private standpoint – a necessary condition;
2 Limited access to these possibilities because of personal qualifications and external circumstances;
3 An economic situation that allows tolerably reliable calculations.

For Schumpeter, and his direct disciples (Freeman, 1979; Kleinknecht, 1981; 1 and Mensch, 1979) innovation is the chief force in what he calls 'economic evolution'. It is worthy of note that for Schumpeter his concept of innovation is broader than some innovation theorists have since argued. For Schumpeter, innovation is not simply the patenting of new inventions, it includes new combinations in organisations, commerce and the market, as well as the creation of new business organisations (Schumpeter, 1961: 66). This, then, may be helpful as we try to consider together the concepts of entrepreneurship and innovation. We also need to recognise a wider definition of innovation that includes more than the hard physical outputs from a traditional science-led institution, such as a technical university (TU) and consider new service offerings as well.

Defining entrepreneurship

For many people the following Innovation in action captures what it is to be an entrepreneur.

Innovation in action

Penny apples – selling them thrice over



Source: Pearson Education Ltd/Westend 61

In his autobiography, the Irish entrepreneur Billy Cullen (2003) tells the story of how, as an eight-year-old boy, he demonstrated sharp entrepreneurial skills. In a poverty-stricken area of Dublin, young Billy would buy wooden crates of apples for a shilling and then sell the apples on a Saturday afternoon to the hundreds of local people who would flock to watch their local football team play. This provided Billy with a healthy profit of a shilling, if he could sell all the apples. But, his entrepreneurial skills did not stop there. He would then take the wooden apple boxes to the football ground and sell them for a penny to people at the back of the crowds, so that they could stand on the box for a better view. And, finally, when the match had finished, Billy would collect up the wooden boxes, break them up and sell them in bundles for firewood.

Entrepreneurship can be described as a process of action that an entrepreneur undertakes to establish an enterprise. Entrepreneurship is a creative activity. It is the ability to create and build something from practically nothing. It is an ability to see an opportunity where others see chaos, contradiction and confusion. Entrepreneurship is an attitude of mind to seek opportunities, take calculated risks and derive benefits by setting up a venture. It comprises numerous activities involved in conception, creation and running an enterprise. Similarly, an entrepreneur is a person who starts such an enterprise. He searches for change and responds to it. There are a wide variety of definitions for an entrepreneur – economists view him as a fourth factor of production along with land labour and capital. Sociologists feel that certain communities and cultures promote entrepreneurship. The USA often is cited as having a culture that supports entrepreneurs. Still others feel that entrepreneurs are innovators who come up with new ideas for products and markets. To put it very simply, an entrepreneur is someone who perceives opportunity, organises resources needed for exploiting that opportunity and exploits it.

Peter Drucker's classic book *Innovation and Entrepreneurship* was first published in 1985 and it was the first book to present innovation and entrepreneurship as a purposeful and systematic activity. According to Drucker, 'Innovation is the specific function of entrepreneurship' and entrepreneurship, 'is the means by which the entrepreneur either creates new wealth-producing resources or endows existing resources with enhanced potential for creating wealth'.

In his book, Drucker focuses on large-scale entrepreneurship, rather than small business management. Drucker's recurring theme is that good entrepreneurship is usually market-focused and market-driven. Contrary to the belief of many, Drucker says that innovation is not inspired by a bright idea, rather it 'is organised, systematic, rational work'. Innovation can be mastered and integrated into a company or non-profit organisation.

In a study of past and future research on the subject of entrepreneurship, Low and MacMillan (1988) define it as 'the process of planning, organising, operating, and assuming the risk of a business venture'. Risk and the willingness to take a risk with one's time and money is surely a key feature of entrepreneurship. It is the analysis of the role of the individual entrepreneur that distinguishes the study of entrepreneurship from that of innovation management. Howard Stevenson, who developed entrepreneurship teaching at Harvard Business School, defines entrepreneurship as follows:

> *Entrepreneurship is the pursuit of opportunity beyond the resources you currently control.*
>
> (Stevenson 1983, 1985, 1990)

This definition takes into account both the individual and the society in which the individual is embedded. The individual identifies an opportunity to be pursued, then, as an entrepreneur, must seek the resources from the broader society. Stevenson argues that entrepreneurship activities can be identified as distinct from that of the administrator (see Table 2.2 below). This is significant, for it recognises that entrepreneurship can be viewed as a mode of management within a corporation.

According to Howard Aldrich (2012), the academic field of entrepreneurship research has grown from groups of isolated scholars doing research on small businesses to an international community of departments, institutes and foundations promoting research on new and high-growth firms. The field is increasingly formalised

Table 2.2 **Definition of entrepreneurship**

A process definition of entrepreneurship		
Key business dimension	**Entrepreneur**	**Administrator**
Strategic orientation	Driven by perception of opportunity	Driven by resources currently controlled
Commitment to opportunity	Quick commitment	Evolutionary with long duration
Commitment process	Multistage with minimal exposure at each stage	Single-stage with complete commitment upon decision
Control of resources	Episodic use of rent of required resources	Ownership or employment of required resources
Management structure	Flat with multiple informal networks	Formalised hierarchy
Reward system	Value-based and team-based	Resource-based individual and promotion oriented

Source: Stevenson (2000).

and anchored in a small set of intellectual bases. Significantly, however, major foundations and many other smaller funding sources have changed the scale and scope of entrepreneurship research today. Virtually all business schools now engage in entrepreneurship research.

Technological entrepreneurship: a question of context

Like so many things in life, our perception of something depends upon our vantage point. For those of us considering entrepreneurship, this is particularly relevant. Do we wish to consider the individual entrepreneur, the organisation, an industry, an economy or even the wider society? Hence, the level of analysis needs to be specified. Also, do we wish to investigate the past or the future? For example, there are many studies that have tried to explain why some firms or individuals were successful – the growth (and fall) of Microsoft or Nokia. There are also historical studies of periods of time, such as the growth of the Roman Empire or the demise of the Ottoman Empire. So the time frame needs to be specified. Finally, do we wish to investigate or study a particular phenomenon or issue? For example, entrepreneurship education has received a great deal of attention from policy makers and politicians as they have tried to enhance their country's economic growth. Nonetheless, this area of business management is problematic. In his book *High-Technology Entrepreneurship*, Professor Ray Oakey provides a comprehensive overview of all aspects of high-technology small firm formation and growth and illustrates that they have not been attractive assessment vehicles for those with money to invest (Oakey, 2012).

Research by Autio et al. (2014) illustrates the importance of context when they compare the attributes of national innovation systems, entrepreneurship and entrepreneurial innovation and its influences on entrepreneurial innovation. The evidence is mounting up all over the world that innovation is key to competitiveness and growth and that entrepreneurial dynamism is key to economic renewal and growth. The focus